

CIRCUITI MAGNETICI

mercoledì 18 maggio 2022 16:21

PROPRIETÀ MAGNETICHE DEI MATERIALI

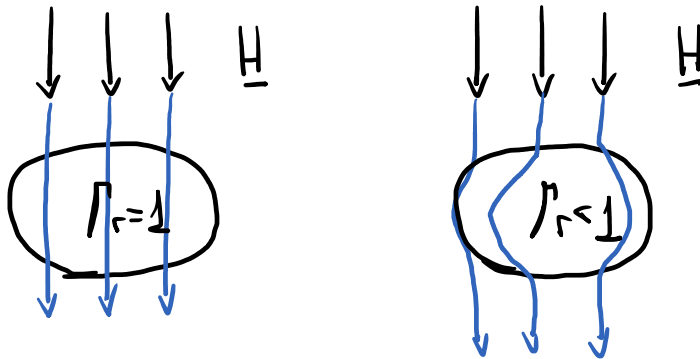
$$\underline{B} = \underline{\mu} \underline{H} = \mu_0 \mu_r \underline{H}$$

$\uparrow$ PERMEABILITÀ MAGNETICA NEL VUOTO
 $\downarrow$ PERMEABILITÀ RELATIVA

LEGGE DI LEGGE
MATERIALI

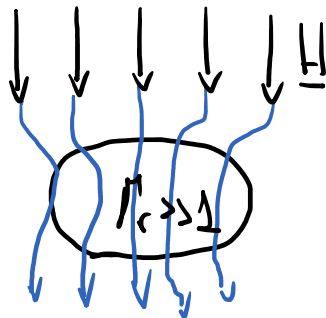
- Classificazione in base a μ_r

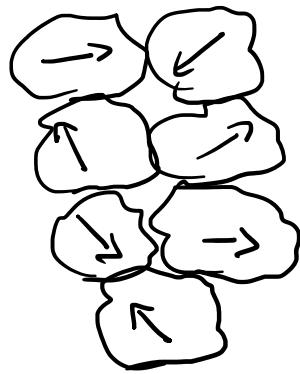
MATERIALI DIAMAGNETICI: $\mu_r < 1$



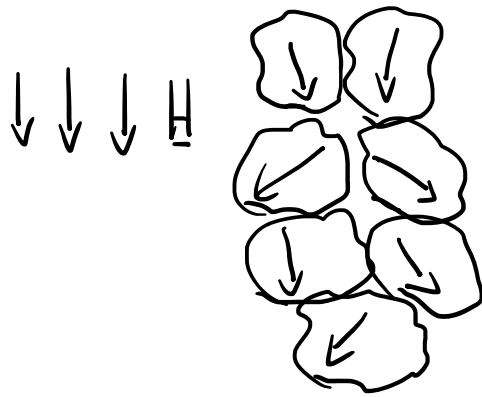
MATERIALI PARAMAGNETICI: $\mu_r > 1$

MATERIALI FERROMAGNETICI: $\mu_r \gg 1$ ($10^3 \div 10^5$)





DOMINI DI WEISS



I DOMINI SI
ORIENTANO PER
EFFETTO DEL CAMPO H

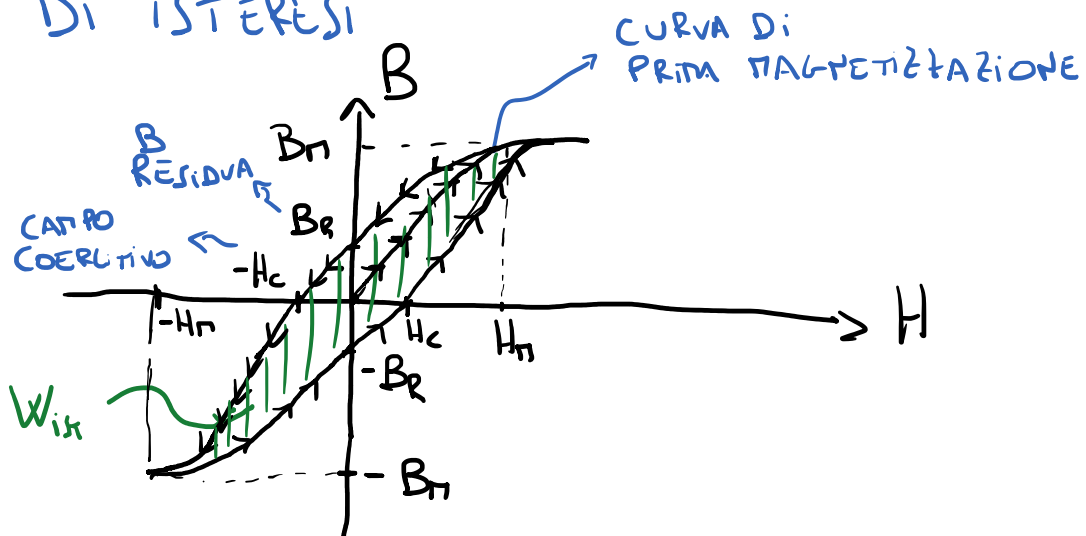
Macchina elettrica: due circuiti elettrici accoppiati
magneticamente

$$V_1 \rightarrow I_1 \xrightarrow{\phi_{H12}=I} H_1 \xrightarrow{B=\mu H} B \xrightarrow{\frac{\partial B}{\partial t} \neq 0} \text{f. em.} \xrightarrow{V_2} I_2$$

$$\underline{B} = \underline{\mu} \underline{H} = \mu_0 (\underline{\mu}_r) \underline{H}$$

$\rightarrow \uparrow \uparrow \Rightarrow$ MATERIALI
FERROMAGNETICI

CICLO DI ISTERESI



$$\frac{1}{f} - B_m$$

PRIMA DELLA SATURATION : $\Gamma_r \gg 1$

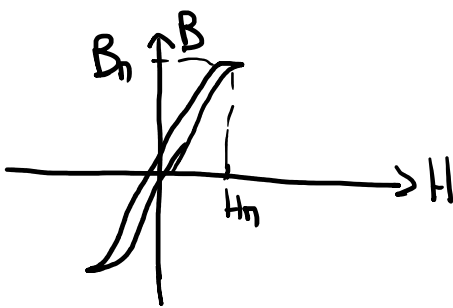
IN SATURATION : $\Gamma_r = 1$

W_{ist} : area del ciclo di isteresi : rappresenta l'energia specifica dissipata in un ciclo per effetto Joule

$P_{ist} = f W_{ist}$: potenza specifica dissipata per effetto Joule $[W/m^3]$

- Classificazione in base al ciclo di isteresi:

DOLCI



$W_{ist} \downarrow \downarrow$

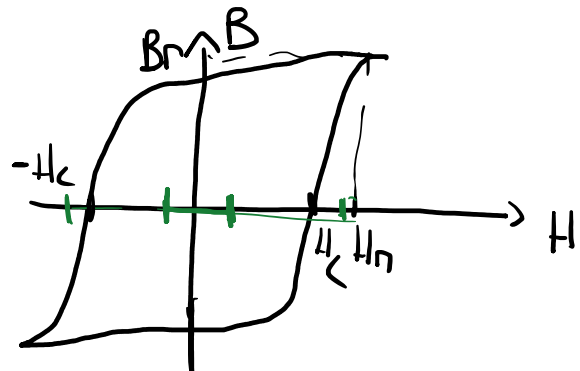
$\Gamma_r \uparrow \uparrow$

$B_m \uparrow$

$H_c \downarrow \downarrow$

Si usano per le macchine elettriche

FORTI



$W_{ist} \uparrow \uparrow$

$B_m \uparrow \uparrow$

$H_c \uparrow \uparrow$

Si usano per i magneti permanenti

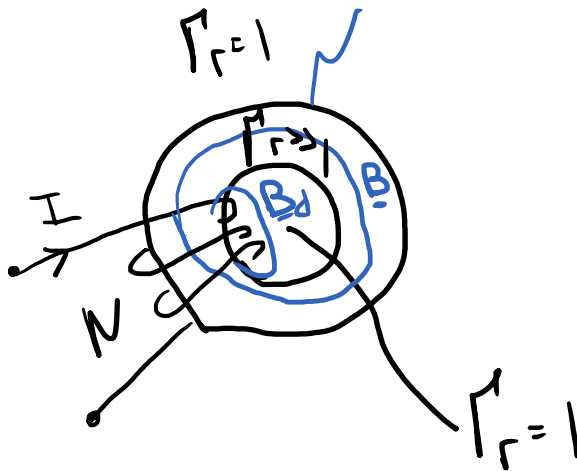
CIRCUITI MAGNETICI

mercoledì 18 maggio 2022 17:11



TUBO DI FLUJO DI B

TOROIDALE



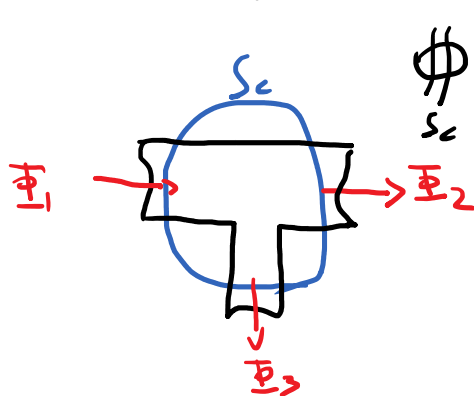
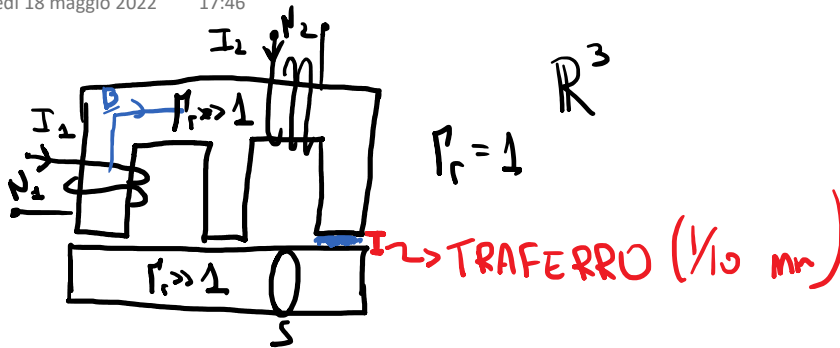
$B_d = B$ disperso

$l_c = I \cdot N$ → NUMERO DI SPIRE

PROBLEMA: il materiale
satura

N_c circuiti elettrici: $\frac{\sigma_{CONDUTTORI}}{\sigma_{ISOLANTE}} = 10^{20}$

N_c circuiti magnetici: $\frac{\mu_{FERR.}}{\mu_{Aria}} = 10^3 \div 10^5$



$$\oint_{S_c} \mathbf{B} \cdot \hat{n} dS = 0 \Rightarrow$$

$$\sum_{k=1}^N \Phi_k = 0$$

LKC MAGNETICA

$$R'_3 = R_3 - \text{avvolgimenti}$$

$$\oint \mathbf{H} \cdot d\mathbf{l} = 0 \rightarrow \mathbf{H} \text{ è conservativo}$$

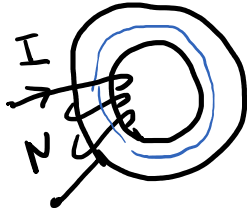
$$\mathbf{H} = -\nabla \Psi_m \quad \xleftrightarrow{\text{ANALOGIA}} \quad (\mathbf{E} = -\nabla V)$$

↓
POTENTIALS
MAGNETICO

$$\sum_{k=1}^N \Psi_k = 0$$

LKT MAGNETICA

CIRCUITI MAGNETICI A PARAM. CONCENTRATI



$$\oint \underline{H} \cdot d\underline{\ell} = I_c = NI$$

$$\iint_S \underline{B} \cdot d\underline{S} = \Phi = \underline{B} \iint_S d\underline{S} = \underline{B} S \rightarrow \underline{B} = \frac{\Phi}{S}$$

$$\underline{B} = \mu \underline{H} \rightarrow \underline{H} = \frac{\underline{B}}{\mu} \rightarrow \oint \frac{\underline{B}}{\mu} d\underline{\ell} = NI \rightarrow$$

$$\rightarrow \oint \frac{\Phi}{\mu S} d\underline{\ell} = NI \rightarrow \frac{\Phi}{\mu S} \oint d\underline{\ell} = NI$$

$$\rightarrow \frac{\Phi}{\mu S} L = NI \rightarrow \underbrace{\frac{L}{\mu S}}_R \Phi = NI$$

$$R = \frac{L}{\mu S} [H^{-1}] \quad \text{RILUTANZA}$$

$$R = \frac{L}{\mu S} [\Omega] \quad \text{RESISTENZA}$$

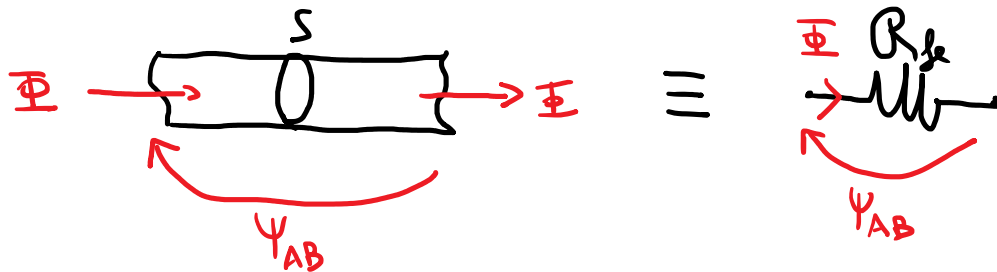
$$NI = [\text{Ampere} \cdot \text{n}^\circ \text{spire}] = [A_{sp}] \quad \begin{array}{l} \text{ΑΠΡΕΡΕΣΠΙΡΕ} \\ \text{FORZA MAGNETO} \\ \text{ΠΟΤΙΣΕ} \\ (\text{gmm}) \end{array}$$

$$\boxed{NI = R \Phi}$$

LEGGE DI OHM
MAGNETICA
(HOPKINSON)

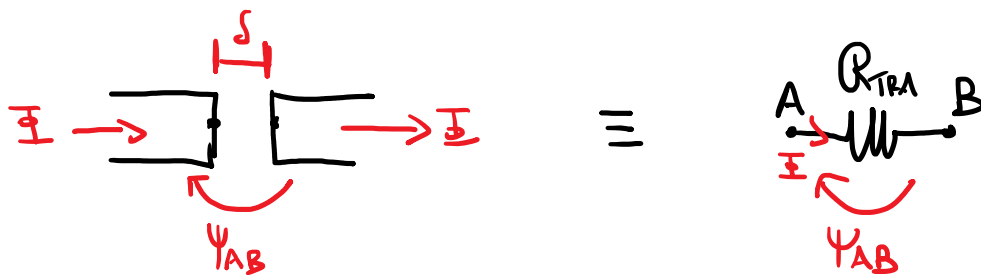
PARAMETRI CONCENTRATI

TRATTO DI MATERIALE FERROMAGN.



$$R_k = \frac{L}{\mu_0 \mu_r S} [H^{-1}]$$

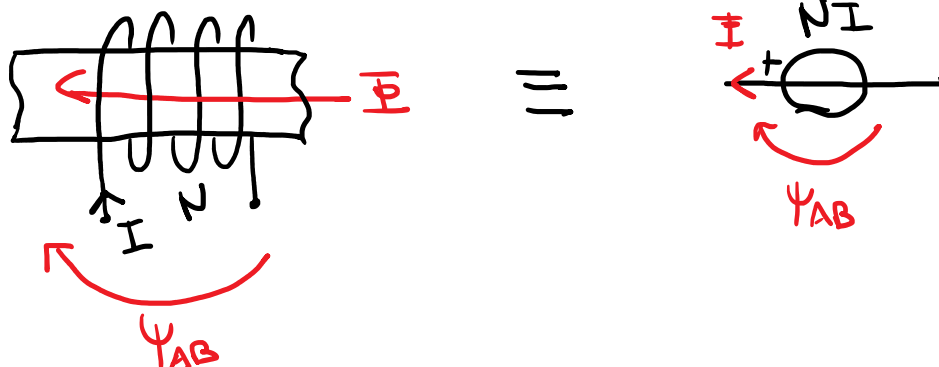
TRASFERRO



$$R_{TRA} = \frac{d}{\mu_0 \cdot S} [H^{-1}]$$

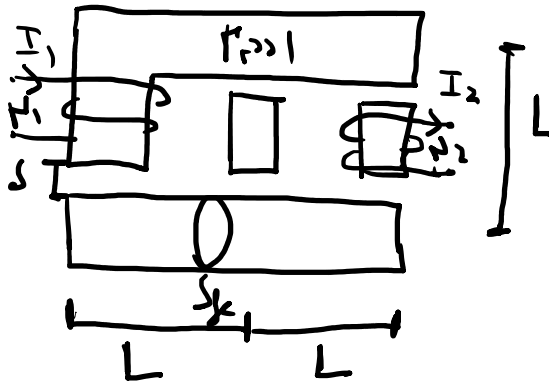


AVVOLGIMENTO



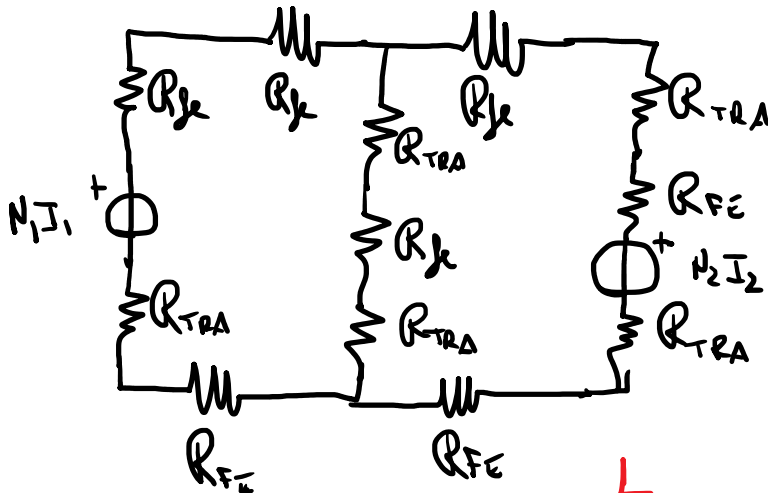
Es.

mercoledì 18 maggio 2022 18:27



$$\begin{aligned} L &= 10 \text{ cm} \\ \delta &= 1 \text{ mm} \\ \mu_{\text{ferro}} &= 3000 \\ S_k &= 300 \text{ cm}^2 \\ N_1 &= 100 \\ I_1 &= 2 \text{ A} \\ N_2 &= 200 \\ I_2 &= 3 \text{ A} \end{aligned}$$

Calcolare i flussi nelle
bombe verticali.

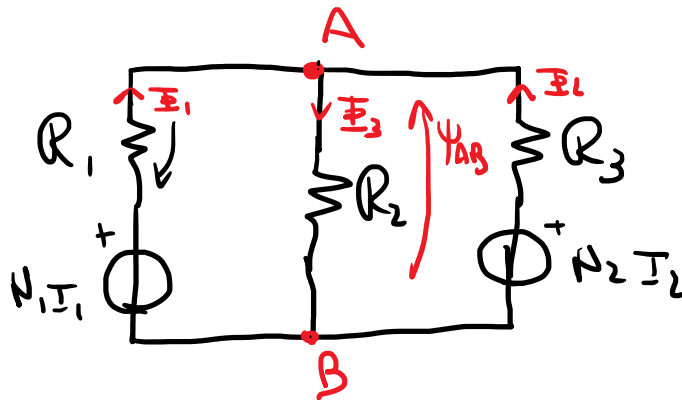


$$R_k = \frac{L}{\mu_0 \cdot \mu_r \cdot S} = \frac{0.1}{1,256 \cdot 10^{-6} \cdot 3000 \cdot 300 \cdot 10^{-4}} = 885 \text{ H}^{-1}$$

$$R_{\text{TRA}} = \frac{\delta}{\mu_0 \cdot 1 \cdot S} = \frac{10^{-3}}{1,256 \cdot 10^{-6} \cdot 1 \cdot 300 \cdot 10^{-4}} = 26539 \text{ H}^{-1}$$

$$I_{mm\ 1} = N_1 I_1 = 200 \text{ A}_{sp}$$

$$I_{mm\ 2} = N_2 I_2 = 600 \text{ A}_{sp}$$



$$R_1 = 29194 \text{ H}^{-1}$$

$$R_2 = 53963 \text{ H}^{-1}$$

$$R_3 = 55733 \text{ H}^{-1}$$

$$\Psi_{AB} = \frac{N_1 I_1 / R_1 + N_2 I_2 / R_2}{1/R_1 + 1/R_2 + 1/R_3}$$

$$\Phi_1 = \frac{N_1 I_1 - \Psi_{AB}}{R_1}$$

$$\Phi_2 = \frac{N_2 I_2 - \Psi_{AB}}{R_2}$$

$$\Phi_3 = \frac{\Psi_{AB}}{R_3}$$

$$KCL: \quad \Phi_1 + \Phi_2 = \Phi_3$$